

6+	30%	20%	15%
Years in Data Science	Marketing Cost Saved	Lead Quality Uplift	Revenue Growth

PROFESSIONAL SUMMARY

Data Scientist II with 6+ years building end-to-end machine learning systems, predictive models, and data-driven decision frameworks across EdTech, travel, and real estate — now transitioning toward Applied AI Systems Engineering. Proven track record in churn prediction, lead scoring, demand forecasting, and LTV modelling that directly drive revenue and retention, combined with hands-on backend engineering (FastAPI, Django, Kubernetes) and emerging agentic AI systems work: designing and deploying an LLM-driven, tool-calling workflow (Claude + Databricks via MCP) that lets CRM stakeholders self-serve root-cause analysis. Strong in A/B experimentation, data engineering, and stakeholder partnership — bringing systems thinking and applied ML judgment to production AI.

PROFESSIONAL EXPERIENCE

- Coursera

Sept 2024 – Present

Data Scientist II — Lifecycle Analytics & ML

 - Developed and deployed churn prediction models (XGBoost/LightGBM) on Databricks, identifying at-risk learners and enabling retention interventions across millions of users
 - Designed and deployed an **agentic AI workflow** integrating Claude with Databricks via the Model Context Protocol (MCP) and a custom-authored domain skill (table schemas, SQL patterns, RCA methodology), enabling CRM/Marketing stakeholders to self-serve root-cause analysis without analyst involvement
 - Built customer segmentation models using K-Means and RFM analysis to personalise CRM campaigns across email, push, and in-app channels
 - Designed and executed A/B tests, holdout experiments, and lift studies using Bayesian and frequentist frameworks for lifecycle campaign optimisation
 - Developed automated ML feature pipelines on Databricks for model scoring and campaign audience generation at scale
 - Partnered with CRM and marketing leadership as a data science thought partner — translating outputs into targeting and channel strategy
- Travelopia

Sept 2022 – Aug 2024

Business Analyst — Customer Insights & ML Analytics

 - Built propensity and lead scoring models (Logistic Regression, Random Forest, XGBoost) reducing marketing costs by 30% and improving lead quality by 20%
 - Developed LTV forecasting and time-series demand models (ARIMA, Prophet) for revenue planning and inventory decisions
 - Built automated data pipelines using Docker and Google Cloud with near-real-time refreshes for ML feature stores and reporting
 - Designed Power BI dashboards integrating multi-source data for cross-brand visibility used by senior leadership weekly
- NoBroker Technologies

Nov 2021 – Aug 2022

Business Analyst — Growth Analytics

 - Delivered growth and funnel analytics identifying high-converting cohorts, driving 15% revenue improvement through data-driven targeting
 - Built user behaviour and segmentation models identifying funnel drop-offs and informing product and marketing interventions
 - Automated reporting pipelines in Python (pandas, NumPy), cutting turnaround by 30%
- Quantbot Securities Private Limited (Personal Project)

Jan 2021 – early 2026

Founder & ML / Backend Engineer

 - Independent personal project (parallel to full-time employment) — built a quantitative copy-trading platform: ML models for trade signal generation, price prediction, and portfolio risk analysis
 - Developed backend in FastAPI and Django, containerised with Docker, orchestrated via Kubernetes with async real-time market API integrations
 - Platform shut down in early 2026 due to regulatory constraints on retail algorithmic copy-trading
- Tech Mahindra

Jul 2021 – Oct 2021

Sr. Business Associate — Analytics & Strategy

 - Supported client-facing data analytics and campaign optimisation projects for enterprise clients

- Managed structural design and modelling for steel infrastructure projects

KEY ACHIEVEMENTS

Agentic AI RCA Tool

Claude + Databricks (MCP) + custom skill authoring — stakeholder self-service root-cause analysis

Lead Scoring

XGBoost model cut spend 30%, raised lead quality 20%

Trade Signal ML

Price prediction & portfolio risk models for a live trading platform

Churn Model

ML early warning system enabling personalised retention at Coursera

LTV Forecasting

Time-series models prioritising acquisition by segment

Segmentation

K-Means + RFM clusters powering personalised campaigns at scale

SKILLS

MACHINE LEARNING

XGBoost

LightGBM

Random Forest

Logistic Regression

K-Means

Scikit-learn

AGENTIC AI & LLM SYSTEMS

Claude

Model Context Protocol (MCP)

Agent Skill Design

Context Engineering

Tool-Calling Workflows

FORECASTING & STATS

ARIMA

Prophet

Time-Series

A/B Testing

Bayesian Stats

Hypothesis Testing

ML APPLICATIONS

Churn Prediction

Lead Scoring

LTV Modelling

Propensity Models

Segmentation

Demand Forecasting

DATA ENGINEERING & LANGUAGES

Python

SQL

NumPy

Pandas

Databricks

BigQuery

Airflow

Docker

Kubernetes

Google Cloud

BI, CRM & BACKEND

Power BI

Looker

Data Studio

Plotly

Seaborn

Braze

Lifecycle Analytics

FastAPI

Django REST

CI/CD

EDUCATION

PGDM — Finance & Analytics | International Management Institute, New Delhi | 2019–2021

B.E. — Civil Engineering | Jadavpur University, Kolkata | 2013–2017